IYMC Qualification Round 2026

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Problem A

Find all real roots of the function

$$g(x) = (x-1)(x+4)(\ln x).$$

Solution: Since $\ln x$ is defined only for $x > 0$, the domain is $x > 0$.

- $x - 1 = 0 \Rightarrow x = 1$
- $x + 4 = 0 \Rightarrow x = -4$, but -4 is not in the domain
- $\ln x = 0 \Rightarrow x = 1$

Therefore, the only real root is

$$\boxed{x = 1}.$$

Problem B

Find all integers n such that

$$2^n + 2^{-n} = \frac{5}{2}.$$

Solution: Let $2^n = y$. Then $2^{-n} = \frac{1}{y}$, so

$$y + \frac{1}{y} = \frac{5}{2}.$$

Multiplying by $2y$:

$$2y^2 - 5y + 2 = 0,$$

$$(2y-1)(y-2) = 0.$$

Thus $y = \frac{1}{2}$ or $y = 2$. Since $y = 2^n$,

$$2^n = 2 \Rightarrow n = 1, \qquad 2^n = \frac{1}{2} \Rightarrow n = -1.$$

Therefore,

$$\boxed{n = -1 \text{ or } n = 1}.$$

Problem C

Determine the numerical value of:

$$\frac{\log_4(64) - (-1)^{2026}}{2} + \sum_{k=1}^{4} \frac{\cos\left(\frac{\pi}{2} + k\pi\right)}{(-1)^{k+1}} + \sqrt{\left(\frac{3}{\sqrt{3}}\right)^2} - 1.$$

Solution:

$$\log_4 64 = 3, \qquad (-1)^{2026} = 1.$$

So the first fraction is

$$\frac{3-1}{2} = 1.$$

Also,

$$\cos\left(\frac{\pi}{2} + k\pi\right) = 0$$

for every integer k, so the sum is 0. Finally,

$$\sqrt{\left(\frac{3}{\sqrt{3}}\right)^2} = \frac{3}{\sqrt{3}} = \sqrt{3}.$$

Therefore,

$$1 + 0 + \sqrt{3} - 1 = \sqrt{3}.$$

$$\boxed{\sqrt{3}}$$

Problem D

Prove that for every positive integer n,

$$1 + 2026 + 2026^2 + \cdots + 2026^{n-1}$$

is divisible by 2027 whenever n is a multiple of 2026.

Solution: Let

$$S = 1 + 2026 + 2026^2 + \cdots + 2026^{n-1}.$$

Since n is a multiple of 2026, n is even. Pair the terms:

$$S = (1 + 2026) + (2026^2 + 2026^3) + \cdots + (2026^{n-2} + 2026^{n-1}).$$

Each pair has a factor 2027:

$$1 + 2026 = 2027,$$

$$2026^{2k} + 2026^{2k+1} = 2026^{2k}(1 + 2026) = 2026^{2k} \cdot 2027.$$

Thus,

$$S = 2027\left(1 + 2026^2 + 2026^4 + \cdots + 2026^{n-2}\right).$$

Therefore, S is divisible by 2027.

$$\boxed{2027 \mid 1 + 2026 + 2026^2 + \cdots + 2026^{n-1}}$$

Problem E

A right triangle has legs of lengths a and b. A circle is inscribed in the triangle. Find the area of the region inside the triangle but outside the circle.

Solution: Let the hypotenuse be

$$c = \sqrt{a^2 + b^2}.$$

The area of the triangle is

$$A_{\triangle} = \frac{ab}{2}.$$

For a right triangle, the inradius is

$$r = \frac{a + b - c}{2}.$$

So the area of the incircle is

$$A_{\text{circle}} = \pi r^2 = \pi \left(\frac{a + b - \sqrt{a^2 + b^2}}{2} \right)^2.$$

The required area is therefore

$$\boxed{A = \frac{ab}{2} - \pi \left(\frac{a + b - \sqrt{a^2 + b^2}}{2} \right)^2}$$

or equivalently,

$$\boxed{A = \frac{2ab - \pi \left(a + b - \sqrt{a^2 + b^2} \right)^2}{4}}.$$